

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command

Assign Static IP Addresses to Five Devices

For a lab/home network, use the private subnet **192.168.1.0/24**.

1. IP Address Assignment

Device	IP Address	Subnet Mask	Default Gateway
Laptop 1	192.168.1.10	255.255.255.0	192.168.1.1
Laptop 2	192.168.1.11	255.255.255.0	192.168.1.1
Laptop 3	192.168.1.12	255.255.255.0	192.168.1.1
Desktop 1	192.168.1.13	255.255.255.0	192.168.1.1
Desktop 2	192.168.1.14	255.255.255.0	192.168.1.1

All five devices are in the same subnet:

Network: 192.168.1.0/24

Usable range: 192.168.1.1 – 192.168.1.254

2. Configure the devices

On Windows:

1. Open **Settings** → **Network & Internet**.
2. Open your Ethernet/Wi-Fi connection.
3. Select **IP assignment** → **Edit**.
4. Choose **Manual** and enable **IPv4**.
5. Enter the appropriate IP address from the table.
6. Set subnet prefix length to **24**.
7. Set gateway to 192.168.1.1.
8. Save the settings.

Repeat this for all five devices, using a **different IP address for each device**.

3. Verify the IP configuration

Open Command Prompt:

ipconfig

For example, Laptop 1 should show:

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

4. Test connectivity with ping

From **Laptop 1**, test the other four devices:

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

A successful result looks like:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Repeat from another device, for example Desktop 1:

ping 192.168.1.10

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.14

5. Expected result

Test	Expected result
Laptop 1 → Laptop 2	Successful
Laptop 1 → Laptop 3	Successful
Laptop 1 → Desktop 1	Successful
Laptop 1 → Desktop 2	Successful
Desktop 1 → Laptop 1	Successful

Conclusion

All five devices have **unique static IP addresses in the same 192.168.1.0/24 subnet**. Successful ping replies confirm that the devices can communicate with each other over the local network.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

DHCP Server Setup — Cisco Packet Tracer

We can simulate a DHCP server using a **Cisco router** in Packet Tracer. The router will automatically assign IP addresses to three PCs.

1. Create the topology

- 1 Router
- 1 Switch
- 3 PCs
- Copper straight-through cables

2. Configure the router

Open the router **CLI** and enter:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0
```

```
ip address 192.168.10.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

3. Create the DHCP pool

Reserve the router address and a few addresses for other devices:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

```
ip dhcp pool HOME-LAN
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

The DHCP server can now assign addresses from:

192.168.10.11 to 192.168.10.254

4. Set the PCs to DHCP

For each PC:

PC → Desktop → IP Configuration → DHCP

Do this for PC1, PC2, and PC3.

They should automatically receive addresses similar to:

Device	IP Address	Subnet Mask	Gateway	Assignment
PC1	192.168.10.11	255.255.255.0	192.168.10.1	DHCP
PC2	192.168.10.12	255.255.255.0	192.168.10.1	DHCP
PC3	192.168.10.13	255.255.255.0	192.168.10.1	DHCP

The exact addresses may vary depending on the order in which the PCs request leases.

5. Verify on each PC

Open:

PC → Desktop → Command Prompt

Run:

```
ipconfig
```

For example, PC1 should show:

IPv4 Address.....: 192.168.10.11

Subnet Mask.....: 255.255.255.0

Default Gateway.....: 192.168.10.1

Check PC2 and PC3 as well.

6. Verify the DHCP leases on the router

On the router CLI:

```
show ip dhcp binding
```

You should see something similar to:

IP address	Client-ID/Hardware address
------------	----------------------------

192.168.10.11	...
---------------	-----

192.168.10.12	...
---------------	-----

192.168.10.13	...
---------------	-----

This confirms that the DHCP server has assigned leases to all three devices.

7. Test connectivity

From PC1:

```
ping 192.168.10.12
```

```
ping 192.168.10.13
```

You should receive:

```
Reply from 192.168.10.12: bytes=32 time<1ms
```

```
Reply from 192.168.10.13: bytes=32 time<1ms
```

Also test the gateway:

```
ping 192.168.10.1
```

Result

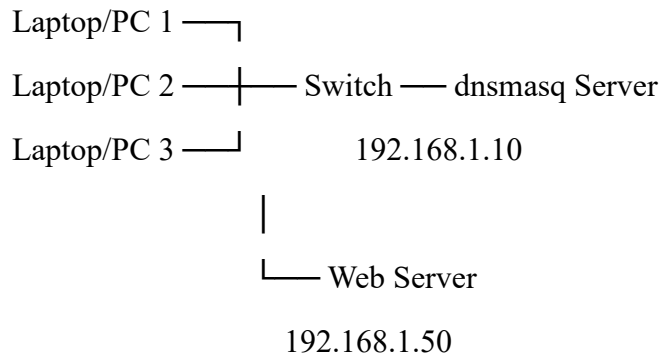
The DHCP server successfully assigned **unique IP addresses** to all three PCs from the configured 192.168.10.0/24 network. No device required manual IP configuration.

3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Configure DNS with dnsmasq for musicplayer.local

This can be simulated on a Linux VM using **dnsmasq**. The custom hostname will point to a local web server at 192.168.1.50.

1. Network setup



Use the same LAN, for example:

Network: 192.168.1.0/24

DNS Server: 192.168.1.10

Web Server: 192.168.1.50

Hostname: musicplayer.local

2. Install dnsmasq

On Ubuntu/Debian:

```
sudo apt update
```

```
sudo apt install dnsmasq
```

3. Configure the custom hostname

Open the dnsmasq configuration:

```
sudo nano /etc/dnsmasq.conf
```

Add:

```
address=/musicplayer.local/192.168.1.50
```

This means:

When a device asks for musicplayer.local, dnsmasq returns 192.168.1.50.

Save the file.

4. Restart dnsmasq

`sudo systemctl restart dnsmasq`

`sudo systemctl status dnsmasq`

The service should show:

Active: active (running)

5. Configure connected devices to use the DNS server

On the router's DHCP settings, set the DNS server to:

192.168.1.10

Alternatively, manually configure the DNS server on each test computer as:

Preferred DNS: 192.168.1.10

6. Test DNS resolution

From a Windows PC:

`nslookup musicplayer.local`

Expected result:

Name: musicplayer.local

Address: 192.168.1.50

On Linux:

`dig musicplayer.local`

The answer should contain:

192.168.1.50

7. Test the local webpage

First verify that the web server works:

`http://192.168.1.50`

Then, from any device connected to the same LAN, open a browser and enter:

`http://musicplayer.local`

The browser should display the webpage hosted at 192.168.1.50.

8. Verify from multiple devices

Device	DNS Query	Expected IP	Result
PC1	musicplayer.local	192.168.1.50	Successful
PC2	musicplayer.local	192.168.1.50	Successful
Laptop	musicplayer.local	192.168.1.50	Successful

Troubleshooting

If it doesn't resolve:

`ipconfig /flushdns`

`nslookup musicplayer.local`

Then check:

1. dnsmasq is running.
 2. Clients are using 192.168.1.10 as their DNS server.
 3. 192.168.1.50 is reachable with ping.
 4. The web server is running.
 5. DNS port **53** is not blocked by a firewall.
 6. Restart dnsmasq after modifying its configuration.
-
4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

VLAN Name	Status	Ports
1 default	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10 sales	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
20 support	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Simulate Internet Access Using NAT in Cisco Packet Tracer

1. Create the topology

Use the following devices:

PC1 —

PC2 — | — Switch — Router — ISP Router — External Server

PC3 —

Example addressing:

Device	Interface	IP Address
PC1	NIC	192.168.1.10/24
PC2	NIC	192.168.1.11/24
PC3	NIC	192.168.1.12/24
NAT Router	G0/0	192.168.1.1/24
NAT Router	G0/1	203.0.113.2/24
ISP Router	G0/0	203.0.113.1/24
ISP Router	G0/1	198.51.100.1/24
External Server	NIC	198.51.100.10/24

Here, 192.168.1.0/24 represents the private LAN and 198.51.100.0/24 represents the simulated Internet.

2. Configure the LAN PCs

For PC1:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC2:

IP Address: 192.168.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC3:

IP Address: 192.168.1.12

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

3. Configure the NAT router

Open the router CLI:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet 0/1

```
ip address 203.0.113.2 255.255.255.0
```

```
ip nat outside
```

```
no shutdown
```

```
exit
```

Create a NAT access list:

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

Enable PAT/NAT overload:

```
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
```

Configure the default route toward the ISP:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

4. Configure the ISP router

On the ISP router:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0
```

```
ip address 203.0.113.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
interface gigabitEthernet 0/1
```

```
ip address 198.51.100.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

Add a route back to the LAN:

```
ip route 192.168.1.0 255.255.255.0 203.0.113.2
```

5. Configure the external server

On the external server:

IP Address: 198.51.100.10

Subnet Mask: 255.255.255.0

Default Gateway: 198.51.100.1

Go to:

Server → Services → HTTP → On

Create a simple webpage:

```
<h1>Simulated Internet</h1>
```

```
<p>NAT configuration is working successfully.</p>
```

6. Test the connection

From **PC1 → Desktop → Command Prompt**, first test the LAN router:

```
ping 192.168.1.1
```

Then test the ISP router:

```
ping 203.0.113.1
```

Finally, test the simulated external server:

```
ping 198.51.100.10
```

Successful replies confirm connectivity.

7. Test using a web browser

On PC1, open **Web Browser** and enter:

```
http://198.51.100.10
```

You should see:

Simulated Internet

NAT configuration is working successfully.

Repeat the test from PC2 and PC3.

8. Verify NAT translations

On the NAT router, run:

show ip nat translations

You should see translations similar to:

Inside local	Inside global
192.168.1.10	203.0.113.2
192.168.1.11	203.0.113.2
192.168.1.12	203.0.113.2

This demonstrates that multiple private LAN devices are sharing the router's external address.

You can also run:

show ip nat statistics

Expected result

PC1 192.168.1.10 —┐

PC2 192.168.1.11 —┴—> NAT Router —> ISP —> 198.51.100.10

PC3 192.168.1.12 —┘

|

Private → Public

NAT/PAT